

BIOL 220 Describing data

'okana

The Moku System: Managing Biocultural Resources for Abundance within Social-Ecological Regions in Hawai'i

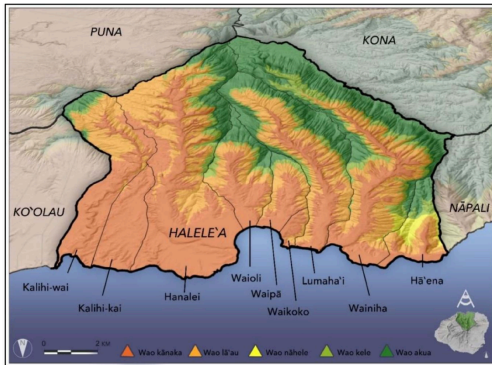


Figure 2. A spatial model depicting the layout for the social-ecological region (*moku*) of Halele'a on the island of Kaua'i, including the social-ecological zones (*wao*) that dictated resource management in each social-ecological community (*ahupua'a*), as determined by Winter and Lucas [16]. Each *wao* is represented by a different color as indicated in the key. This *moku* contains nine *ahupua'a*, each of which are labeled here by name. Not all *ahupua'a* modeled here have all five *wao* documented from the island of Kaua'i, which indicates that each *ahupua'a* had varying levels of access to and amounts of biocultural resources.

How do we describe **data**?

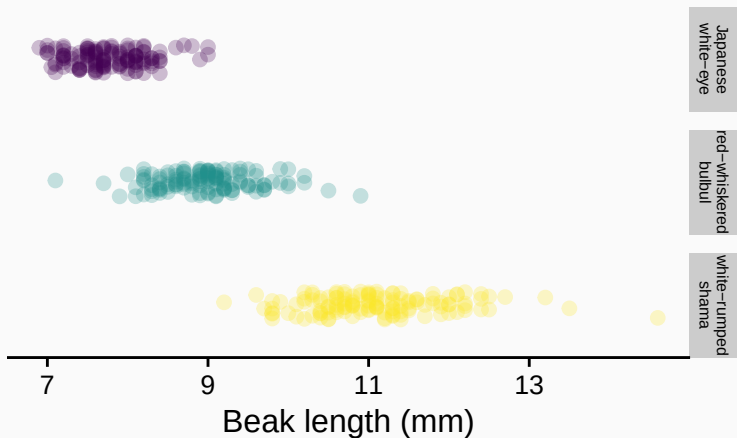
How do we describe **data**?

- Why do we make the choices we make in describing data?
- These choices get bundled into knowledge system
- Whose knowledge system(s) do we learn and why?

Learning Objectives

1. Describe the *location* of numerical variables using the **mean** and **median**
2. Describe the *spread* of numerical variables using the **standard deviation**, **variance**, **coefficient of variation**, and **interquartile range**
3. Describe categorical variables using the **proportion**
4. Know when to use different descriptive statistics

How do bird beaks differ?



Data from Gleditsch, J. M., & Sperry, J. H. (2020). *Ecology and Evolution*, 10(21), 12157-12169.

Describing location and spread of numerical data

Japanese white-eye



red-whiskered bulbul



white-rumped shama



Images from Wikipedia

Describing location and spread of numerical data

the Mean describes location

Descriptor	Parameter	Estimate	Units
mean	μ	$\bar{Y}$	measurement unit (e.g. cm, seconds, etc.)

Notes:

- μ is Greek letter pronounced like “mew”
- $\bar{Y}$ pronounced like “Y bar”

Sample mean

In words:

*The **sample mean** is the sum of all the observations in a sample divided by n , the number of observations*

In math:

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

Sample mean

A subset of 8 white-rumped shama beak length measurements



Beak lengths in mm

10.8

9.2

12.0

11.4

12.1

10.6

10.6

11.6

Data from Gleditsch, J. M., & Sperry, J. H. (2020). *Ecology and Evolution*, 10(21), 12157-12169.

Sample mean

Pause & Think

- What is n ?
- What is $\sum_{i=1}^n Y_i$?
- What is $\bar{Y}$?

i	Y_i
1	10.8
2	9.2
3	12.0
4	11.4
5	12.1
6	10.6
7	10.6
8	11.6

Describing location and spread of numerical data

the Standard Deviation describes spread

Descriptor	Parameter	Estimate	Units
mean	μ	$\bar{Y}$	measurement unit
standard deviation	σ	s	measurement unit

Notes:

- σ (“sig-ma”) is Greek letter “s”

Sample standard deviation

In words:

*The **standard deviation** is a common measure of the spread of data. It indicates how far the different measurements are (on average) from the mean.*

In math:

$$s^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1}$$
$$s = \sqrt{s^2}$$

s^2 is the sample variance

Take square root to get standard deviation: $\sqrt{s^2} = s$

FAQ

Q1: Why does the sum of squared deviations $(\sum_{i=1}^n (Y_i - \bar{Y})^2)$ appear in the numerator?

A1: Negative and positive deviations have the same effect on average deviation. For example, suppose $\bar{Y} = 0$:

$$(1 - 0)^2 = (1)^2 = 1$$

$$(-1 - 0)^2 = (-1)^2 = 1$$

FAQ

Q2: Why use the sum of squared deviations rather than absolute deviations?

A2: The squared values turn out to have better statistical properties than absolute values.

FAQ

Q3: Why is it divided by $n - 1$ rather than n ?

A3: Dividing by n is **biased** and *underestimates* the population standard deviation. Matters most when n is small.

Sample standard deviation

A subset of 8 white-rumped shama beak length measurements



Beak lengths in mm

10.8

9.2

12.0

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11.6

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Sample standard deviation

Remember: $\bar{Y} = 11.04$ mm

i	Y_i	Deviations ($Y_i - \bar{Y}$)
1	10.8	$10.8 - 11.04 = -0.24$
2	9.2	
3	12.0	
4	11.4	
5	12.1	
6	10.6	
7	10.6	
8	11.6	
Sum	88.3	

Sample standard deviation

By definition, sum of deviations equals 0

i	Y_i	Deviations ($Y_i - \bar{Y}$)
1	10.8	-0.24
2	9.2	-1.84
3	12.0	0.96
4	11.4	0.36
5	12.1	1.06
6	10.6	-0.44
7	10.6	-0.44
8	11.6	0.56
Sum	88.3	0.00

Sample standard deviation

Sum of squared deviations is always positive

i	Y_i	Deviations $(Y_i - \bar{Y})$	Squared deviations $(Y_i - \bar{Y})^2$
1	10.8	-0.24	$(-0.24)^2 = 0.0576$
2	9.2	-1.84	
3	12.0	0.96	
4	11.4	0.36	
5	12.1	1.06	
6	10.6	-0.44	
7	10.6	-0.44	
8	11.6	0.56	
Sum	88.3	0.00	

Sample standard deviation

Sum of squared deviations is always positive

i	Y_i	Deviations ($Y_i - \bar{Y}$)	Squared deviations ($Y_i - \bar{Y}$) ²
1	10.8	-0.24	0.0576
2	9.2	-1.84	3.3856
3	12.0	0.96	0.9216
4	11.4	0.36	0.1296
5	12.1	1.06	1.1236
6	10.6	-0.44	0.1936
7	10.6	-0.44	0.1936
8	11.6	0.56	0.3136
Sum	88.3	0.00	6.3188

Sample standard deviation

$$s^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1} = \frac{6.3188}{7} = 0.9027 \text{ mm}^2$$

$$s = \sqrt{0.9027} = 0.9501 \text{ mm}$$

The standard deviation has the same units as the measurements (mm in this case), so you can describe the departure from the mean in a consistent way.

“Almost 95% of data typically fall within two standard deviations of the mean”

Describing location and spread of numerical data

Coefficient of Variation is standard deviation as a percentage of the mean

Descriptor	Parameter	Estimate	Units
mean	μ	$\bar{Y}$	measurement unit
standard deviation	σ	s	measurement unit
variance	σ^2	s	measurement unit ²
coefficient of variation	$\sigma/\mu \times 100\%$	CV	percent (%)

Notes:

- CV is useful for comparing data with different measurement units

Coefficient of variation

In words:

*The **coefficient of variation** is the standard deviation expressed as a percentage of the mean*

In math:

$$CV = \frac{s}{\bar{Y}} \times 100\%$$

Coefficient of variation

A subset of 8 white-rumped shama beak length measurements



Beak lengths in mm

10.8

9.2

12.0

11.4

12.1

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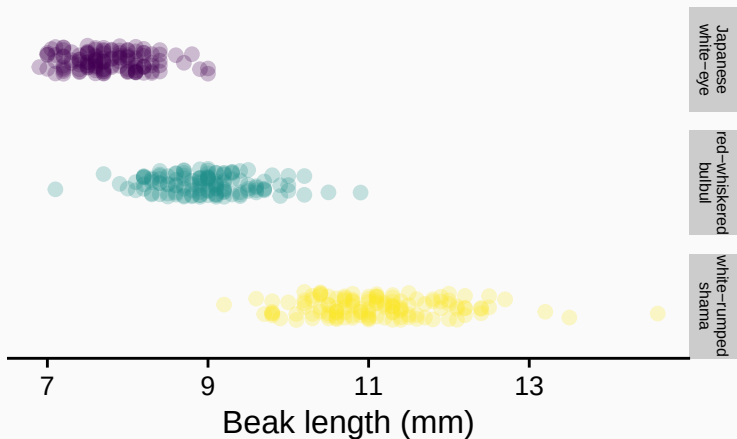
Coefficient of variation

$$CV = \frac{s}{\bar{Y}} \times 100\% = \frac{0.9501}{11.04} \times 100\% = 8.61\%$$

Describing location and spread of numerical data

- Rank species by mean beak length by eye
- Rank species by standard deviation beak length by eye

How do bird beaks differ?



Describing location and spread of numerical data

- Rank species by mean beak length by eye
- Rank species by standard deviation beak length by eye

Species	$\bar{Y}$	s	CV
Japanese white-eye	7.76	0.45	5.74%
red-whiskered bulbul	8.97	0.57	6.35%
white-rumped shama	11.09	0.83	7.44%

Note: statistics for larger data set (not subset of 8)

The mean and the spread increase together. This is common for morphological data.

Describing location and spread of numerical data

Descriptor	Parameter	Estimate	Units
median	Median	M	measurement unit

Notes:

- “median is the middle observation in a set of data”

If n is *odd*:

$$\text{Median} = Y_{([n+1]/2)}$$

If n is *even*:

$$\text{Median} = [Y_{(n/2)} + Y_{(n/2+1)}]/2$$

Median

Median



9.2 10.6 10.6 10.8 11.4 11.6 12 12.1

$$n = 8$$

$$\text{Median} = \frac{Y_4 + Y_5}{2} = \frac{10.8 + 11.4}{2} = 11.10 \text{ mm}$$

Describing location and spread of numerical data

Descriptor	Parameter	Estimate	Units
median	Median	M	measurement unit
interquartile range	Interquartile range	IQR	measurement unit

Notes:

- “ IQR is the difference between the third and first quartiles of the data. It is the span of the middle 50% of the data.”

Interquartile range

First
quartile

Median



9.2

10.6

10.6

10.8

11.4

11.6

12

12.1

$$j = 0.25n = 0.25 \times 8 = 2$$

$$\text{First quartile} = \frac{Y_2 + Y_3}{2} = \frac{10.6 + 10.6}{2} = 10.60 \text{ mm}$$

Interquartile range

First
quartile

Median

Third
quartile



9.2

10.6

10.6

10.8

11.4

11.6

12

12.1

$$k = 0.75n = 0.75 \times 8 = 6$$

$$\text{Third quartile} = \frac{Y_6 + Y_7}{2} = \frac{11.6 + 12}{2} = 11.80 \text{ mm}$$

Interquartile range

Interquartile range = Third quartile – First quartile = 1.20 mm

$$IQR = 11.80 - 10.60 = 1.20 \text{ mm}$$

3. Describe categorical variables using the **proportion**

Descriptor	Parameter	Estimate	Units
proportion	proportion	$\hat{p}$	probability

Notes:

- the **sample proportion** is similar to the **sample mean**

Sample proportion

In words:

*The **sample proportion** is a measure of the **location** of categorical data. It indicates how frequently a specific category is found in the data.*

In math:

$$\hat{p} = \frac{N_{[Y=\text{category}]}}{n}$$

Sample proportion

Heads or tails?

H T T T H H T H T T



Sample proportion

Heads or tails?

H T T T H H T H T T

$$N_{[Y=\text{tails}]} = 6$$

$$n = 10$$

$$\hat{p}_{\text{tails}} = 0.6$$

Edith Kanaka'ole



E hō mai ka 'ike mai luna mai ē
O nā mea huna no'eau o nā mele ē
E hō mai
E hō mai
E hō mai ē

source: kaainamomona.org

4. Know when to use different descriptive statistics

When data are roughly symmetrical and there are not many extreme values

- mean, standard deviation, variance are commonly used
- CV if you want to compare values on different scales

When data are skewed and/or there are many extreme values

- median tells more about a typical unit in the population and is less sensitive to extreme values
- IQR is less sensitive to extreme values than standard deviation

Rounding

- No standard rules
- For this class, round to 2 decimal places